PalmSource™ 99

Adding Communications to Your App (TCP/IP)

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Introduction

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- Designing your app to communicate
- Pitfalls of communications
- Being Palm™ device-friendly
- Being responsive
- NetLib APIs

Designing Your App to Communicate

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- Bolting Comms onto an existing app
- Porting from another platform
- Instead, get the UI and the Comms to work hand in hand

Pitfalls of Communications

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- Blocking and the perils therein
- Being responsive to the user
- Being responsive to the other side

What Is Blocking

- Long reads or writes without timeouts
- Stops the UI in its tracks
- Prevents the app from responding to errors
- The user can’t interact either

Palm™ Device-Friendly Communications

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- Keep the Zen of Palm uppermost
- Design around presenting the data
- Keep the communications “in the background”

Designing around the Data

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- Presenting the data
- Manipulating the data
- Sending and receiving are usually secondary

Backgrounding the Communications

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- Work with the event loop
- Be quick about it!
- Don’t wait
- Don’t make them wait

Work with the Event Loop

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- Short timeouts
- Polling communications

Be Quick about It

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- Get what's available then get out
- Always check what's available and what can be sent
- Keep the incoming buffer as empty as you can
- Keep the outgoing buffer as full as you can

Example of Event Loop with Comms Embedded

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```
EventType   event;
SWord numReady;
NetFDSetType readFds;
NetFDSetType writeFds;
NetFDSetType exceptFds;
Word maxFd;
Err err;
```

Example of Event Loop with Comms Embedded (Cont.)

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```
do {
   netFDZero(&readFDs);
   netFDZero(&writeFDs);
   netFDZero(&exceptFDs);
   netFDSet(sysFileDescStdIn, &readFDs);
   netFDSet(socketRef, &readFDs);
   if (sendPending) {
      netFDSet(socketRef, &writeFDs);
   }
   numReady = NetLibSelect(refnum, maxFd,
      &readFds, &writeFds, &exceptFds, 100,
      &err);
```

Example of Event Loop with Comms Embedded (Cont.)

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```
if (numReady < 0) {
   // Handle a select error

   continue; // or break, whatever's appropriate
}
UIReady = netFDIsSet(sysFileDescStdIn, &readFds);
readReady = netFDIsSet(socketRef, &readFds);
sendReady = netFDIsSet(socketRef, &writeFds);
```

Example of Event Loop with Comms Embedded (Cont.)

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```
    if ((numReady == 0) || (UIReady)) {
       EvtGetEvent(&event, 10);
       HandleEvent(&event);
    }
    if (readReady) {
       ReadSome(socketRef);

       HandleInput();
    }
    if (sendPending && sendReady) {
       SendSome();
    }
}
while (event.eType != appStopEvent);
```

Example of Event Loop with Comms Embedded (Cont.)

- ReadSome uses NetLibReceivePB
- WriteSome uses NetLibSendPB
- HandleInput knows that everything expected may not have arrived yet

NetLib API Overview

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- Not exhaustive
- Focused on performance
- Focused on responsiveness

NetLib API Overview (Cont.)

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- NetLibDmReceive
  - Particularly useful when memory is tight
  - Use the timeout
  - Be sure at least one byte is ready before calling (use select)
  - Check the error result!

NetLib API Overview (Cont.)

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- NetLibReceivePB
  - Gives you what's been received so far into your own buffer
  - Best way to keep the input buffers clear
  - Always check how many bytes were actually received
  - Use the timeout!
  - Check the error result!

NetLib API Overview (Cont.)

- NetLibSendPB
  - Sends from your own buffer
  - Always check how many bytes were actually sent
  - Use the timeout!
  - Check the error result!
  - Am I getting redundently repetitive?

NetLib API Overview (Cont.)

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- NetLibSelect
  - Build your event loop around select using the StdIn descriptor
  - Use the timeout
  - Be wary of a couple of bugs/misfeatures
    - Remote close
    - Connection to a listening socket

NetLib API Overview (Cont.)

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- NetLibScoketOptionSet
  - TCPNoDelay
  - SockNonBlocking

NetLib API Overview (Cont.)

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- NetLibSocketShutdown
  - May release unnecessary buffer space
  - Better: Can result in better performance

NetLib API Overview (Cont.)

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- NetLibSend
- NetLibSendPB
- NetLibSocketAccept
- NetLibSocketBind
- NetLibSocketConnect
- NetLibSocketListen
- NetLibSocketOpen
- NetLibSocketShutdown

Keeping Things Responsive

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- Keep the event loop running!
- Use short time outs
- Only read when you know something is available


Q&A


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